
RESIDUAL-STREAM COVARIANCE GEOMETRY IN TRANSFORMER LANGUAGE MODELS

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ABSTRACT

Covariance spectra are widely used to summarize language-model representations, but in pre-normalized transformers the stored raw residual stream is not the same object consumed by attention and MLP sublayers. This mismatch leaves an important interpretive gap: an apparently low-dimensional raw residual state may be mistaken for low-dimensional computation. This paper studies that gap through a frozen-activation analysis of residual-stream covariance in GPT-2, Pythia, Qwen2.5, and Mixtral on WikiText-103. The main result is that Qwen2.5 and Mixtral exhibit sustained near-rank-1 raw residual covariance across many middle layers, with highly aligned leading principal directions, whereas GPT-2 and Pythia do not show the same plateau. However, RMSNorm and block-stage diagnostics show that the dominant raw direction is strongly attenuated in normalized sublayer inputs, even when the direction reappears in later raw residual states. These findings indicate that residual covariance is an important diagnostic of storage geometry, but not a direct proxy for computation-facing dimensionality.

1 Introduction

Transformer language models build representations through residual addition, attention, feed-forward computation, and normalization [1]. The *raw residual stream* refers here to the unnormalized hidden state stored between blocks before RMSNorm or LayerNorm is applied. Although this state has a high nominal hidden dimension, its covariance across tokens can concentrate in a much smaller effective subspace at particular depths.

Residual covariance matters because it summarizes how token states occupy the shared communication pathway used throughout a transformer. A highly concentrated spectrum can indicate a persistent common direction, a storage convention, or a bottleneck-like representation, so it is a useful first diagnostic for mechanistic interpretability and representation analysis. At the same time, covariance can be misleading if the measured object is not the object consumed by downstream computation.

Prior work has shown that language-model representations are often anisotropic and contain dominant shared components [2, 3]. A common way to summarize such internal representations is to measure covariance spectra across hidden states. However, in pre-normalized transformer architectures, the raw residual stream is not necessarily the same object consumed by attention and MLP sublayers. This creates an interpretive risk: a raw residual state may appear low-dimensional even when the computation-facing normalized input remains high-dimensional. Existing anisotropy and effective-rank analyses establish that dominant directions occur, but they do not by themselves resolve whether raw residual covariance collapse survives the normalization boundary inside modern pre-normalized LLM blocks.

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The novelty of this study is to separate raw residual storage geometry from normalization-facing computation geometry under one fixed measurement protocol. The analysis asks whether low-dimensional residual covariance reflects a collapse of computational state, or whether it is partly a property of the raw storage space that is altered before sublayer computation. Pretrained language models are analyzed without changing model weights or activations. Layerwise covariance spectra, effective dimensionality, and dominant-direction statistics are measured across several model families, then raw residual states are compared with normalized views and intermediate block stages.

The main contributions are:

- Sustained near-rank-1 raw residual covariance regimes are identified in Qwen2.5 and Mixtral, but not in GPT-2 or Pythia under the same protocol.
- The dominant raw principal direction is shown to be aligned across many sampled layers, indicating a persistent common direction in the raw residual stream.
- Normalization ablations and block-stage diagnostics show that normalized sublayer inputs remain substantially higher-dimensional than the corresponding raw residual covariance.
- A token-budget sensitivity check shows that the Qwen2.5-7B near-rank-1 interval is unchanged across 75k, 150k, and 300k requested-token sweeps.
- Residual covariance geometry is argued to require separation from computation-facing geometry when interpreting transformer internals.

2 Related Work

Residual-stream geometry and mechanistic interpretability. The residual stream is the shared pathway through which transformer blocks read and write token states [1]. Mechanistic-interpretability work has studied circuit-level structure such as induction heads and automated circuit discovery [4, 5], while more residual-stream-specific work shows that belief-state information and sparse-autoencoder features can persist, split, or reappear across layers [6, 7]. Sparse feature methods and covariance PCs are therefore complementary views of the same residual space: sparse features aim for interpretable directions, while covariance spectra summarize how token states occupy the ambient space. The present analysis differs from these studies by focusing on raw residual covariance across frozen pretrained models and by explicitly comparing it with normalization-facing sublayer inputs.

Representation geometry and normalized views. Language representations are often anisotropic: word and contextual embeddings can contain dominant common components, and whitening or component-removal methods can change downstream representation quality [2, 3, 8, 9]. Effective-rank and participation-ratio diagnostics provide label-free summaries of such spectra [10, 11]. This literature motivates the spectral measurements in this paper but also cautions against equating low-rank covariance with low-dimensional computation. The added contribution here is to pair raw covariance spectra with normalized views and block-stage measurements, which separates apparent raw residual covariance collapse from computation-facing geometry.

Normalization and model-family controls. Normalization placement affects transformer optimization and the geometry seen by sublayers [12, 13], and RMSNorm is now common in decoder-only LLMs [14]. This makes Qwen2.5 and Mixtral useful targets for comparing raw residual states with RMSNorm-normalized inputs, while GPT-2 and Pythia provide dense LayerNorm-based contrast cases [15, 16, 17, 18]. Mixtral also motivates aggregate routing checks because MoE routing behavior need not directly correspond to human-interpretable expert specialization [19, 20]. Router load and entropy are therefore reported alongside residual-stream spectra, but interpreted only as aggregate controls rather than semantic expert labels.

3 Experimental Methodology

The internal geometry of pretrained transformer language models is studied by measuring residual-stream activations across depth. All models are evaluated in a frozen setting: no training, fine-tuning, weight updates, or activation interventions are performed. The goal is to characterize how token representations are distributed throughout residual-stream feature space across layers and model families.

Figure 1 summarizes the full measurement pipeline, from WikiText-103 tokens through frozen-model activation capture, centered covariance estimation, eigendecomposition, and geometry-metric computation.

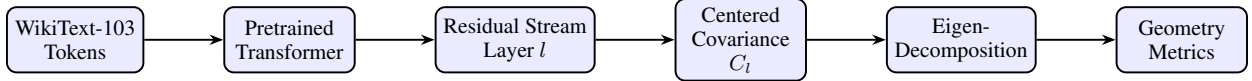


Figure 1: Measurement pipeline for residual-stream geometry analysis.

3.1 Models

The evaluated suite consists of GPT-2 [15], Pythia [16], Qwen2.5 [17], and Mixtral-8x7B [18]. The suite spans model scale, architecture family, instruction tuning, and mixture-of-experts (MoE) routing behavior. Table 1 summarizes the models and their role in the analysis.

Table 1: Model suite used in the residual-geometry analysis.

Model	Size	Architecture	Role
GPT-2 small	124M	Dense, LayerNorm	Small dense baseline
GPT-2 medium	355M	Dense, LayerNorm	Same-family scale comparison
Pythia-1.4B	1.4B	Dense, LayerNorm	Intermediate dense baseline
Qwen2.5-7B	7B	Dense, RMSNorm	Modern dense base model
Qwen2.5-7B-Instruct	7B	Dense, RMSNorm	Matched instruction-tuned 7B model
Qwen2.5-14B	14B	Dense, RMSNorm	Larger dense base model
Qwen2.5-14B-Instruct	14B	Dense, RMSNorm	Matched instruction-tuned 14B model
Mixtral-8x7B	8x7B	Sparse MoE, RMSNorm	Sparse MoE comparison

3.2 Dataset and Tokenization

All models are evaluated on the Salesforce WikiText-103 validation split [21]. WikiText-103 is useful for this purpose because it provides long, natural-language passages with minimal task formatting, allowing residual covariance to be measured on ordinary autoregressive text rather than on short prompts, labels, or instruction templates. Each model uses its native tokenizer, with tokenizer-added special tokens disabled. Because tokenization differs across model families, cross-model comparisons should be interpreted at the level of model-specific token representations rather than identical token sequences.

The main residual-stream sweep uses contiguous, unpadded 512-token chunks, so no padding tokens are included in covariance estimates. A 512-token window is long enough to sample within-document context and many token positions per forward pass, while keeping all evaluated models within available GPU memory for full-depth activation capture. It also avoids mixing sequence-length effects with architecture effects; increasing the context length would change both the computational cost and the distribution of token positions being summarized. Token budgets differ because of memory and compute constraints for larger models; all covariance estimates are computed from the full observed token set for each model, and comparisons are interpreted as protocol-level measurements rather than exact sample-size-controlled estimates. Forward passes run on CUDA with bfloat16 model activations where supported; covariance statistics are accumulated in float32. Table 2 reports the exact sampling protocol.

Table 2: Main residual-stream sampling protocol. Sequence length is 512 for all rows.

Model	Token budget	Observed tokens	512-token chunks	Batch
GPT-2 small	300k	242176	473	8
GPT-2 medium	300k	242176	473	4
Pythia-1.4B	250k	247296	483	2
Qwen2.5-7B Base	200k	199680	390	1
Qwen2.5-7B Inst.	200k	199680	390	1
Qwen2.5-14B Base	150k	149504	292	1
Qwen2.5-14B Inst.	150k	149504	292	1
Mixtral-8x7B	75k	74752	146	1

3.3 Token-Budget Sensitivity and Computational Cost

The unequal token budgets in Table 2 are a possible confound, so an additional Qwen2.5-7B base sensitivity sweep was run at 75k, 150k, and 300k requested tokens. Because only complete 512-token chunks are used and the validation stream is finite, the observed counts were 74,752, 149,504, and 259,584 tokens. The near-rank-1 interval was unchanged

across all three budgets: layers 4–17 satisfy the criterion from Section 3.4.1 in every run. The minimum normalized participation-ratio dimension remained at layer 5 with $D_{\text{PR}}/d = 0.000280$, and the maximum top-1 variance share changed only from 0.997365 at 75k to 0.997437 at the largest run. This sensitivity check indicates that the Qwen2.5-7B covariance-spectrum conclusion is not an artifact of the original token budget.

Table 3: Qwen2.5-7B token-budget sensitivity.

Requested tokens	Observed tokens	Runtime	Min D_{PR}/d	Near-rank-1 layers
75k	74752	206.4 s	0.000280	4–17
150k	149504	357.3 s	0.000280	4–17
300k	259584	607.0 s	0.000280	4–17

The computational cost is dominated by frozen forward passes and online covariance accumulation. Raw activations are not stored to disk. For each layer, the online estimator maintains a mean vector and a $d \times d$ float32 second-moment matrix, giving $O(Ld^2)$ covariance memory for L captured layers and hidden width d , plus the model weights and one activation batch. Each token contributes an outer-product update, so the covariance accumulation cost scales as $O(NLd^2)$ for N observed tokens. On the recorded CUDA runs, residual-stream sweep runtimes ranged from 8.9 seconds for GPT-2 Small to 27.8 minutes for Qwen2.5-14B Instruct and 12.8 minutes for Mixtral-8x7B. Persistent storage is modest because only metrics CSV files, per-layer eigenvalue arrays, diagnostic CSV files, and figures are saved; raw token-by-feature activation tensors are discarded after their online covariance contribution.

3.4 Residual-Stream Geometry Analysis

For each model, the residual stream entering every transformer block is captured. Each token position across all chunks is treated as one sample. Let

$$X_l \in \mathbb{R}^{n \times d} \quad (1)$$

denote the token-by-feature activation matrix at layer l , where n is the number of observed tokens and d is the hidden dimension.

Activations are centered across tokens,

$$\tilde{X}_l = X_l - \mu_l, \quad (2)$$

where μ_l is the token-wise mean activation vector, and the empirical covariance matrix is computed as

$$C_l = \frac{1}{n-1} \tilde{X}_l^\top \tilde{X}_l. \quad (3)$$

Centering removes the token-wise mean activation so that the spectrum reflects variance around the empirical mean rather than the mean vector itself.

C_l is eigendecomposed to obtain the covariance spectrum

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d. \quad (4)$$

To avoid storing all activations, covariance statistics are accumulated online during dataset traversal.

3.4.1 Geometry Metrics

The analysis uses standard spectral effective-dimensionality diagnostics, including participation-ratio dimension and entropy effective rank [10, 11]. The primary effective-dimensionality measure is

$$D_{\text{PR}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}, \quad (5)$$

which is reported in normalized form as D_{PR}/d .

Entropy effective rank is additionally computed as

$$D_{\text{entropy}} = \exp\left(-\sum_i p_i \log p_i\right), \quad \text{where } p_i = \frac{\lambda_i}{\sum_j \lambda_j}. \quad (6)$$

Spectral concentration is characterized by:

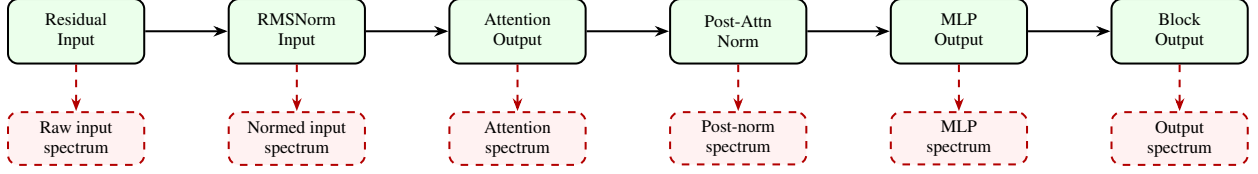


Figure 2: Block-stage diagnostic pipeline. Activations are collected at the residual input, RMSNorm-normalized input, attention output, post-attention normalization, MLP output, and final block output. The same spectral metrics are computed at each stage to compare how low-dimensional raw residual covariance varies across the block.

- Top-1 variance share,
- Top-10 variance share,
- Top-32 variance share,
- k_{90} : number of principal components explaining 90% of variance,
- k_{95} : number of principal components explaining 95% of variance.

k_{90} is the smallest k such that $\sum_{i=1}^k \lambda_i / \sum_j \lambda_j \geq 0.90$, and k_{95} is defined analogously. Several spectral summaries are reported because a single scalar effective-rank measure can obscure whether low dimension comes from one dominant component or from a broader low-rank subspace. The term *near-rank-1 covariance* is used as an operational label when the top principal component explains at least 99% of centered variance and $k_{90} \leq 1$; this is not a claim that the model’s computation is mathematically rank-1.

Finally, feature-subspace novelty between consecutive layers is computed using the leading $k = 32$ principal directions. If $V_l, V_{l+1} \in \mathbb{R}^{d \times k}$ are the two orthonormal bases and σ_i are the singular values of $V_l^\top V_{l+1}$, novelty is

$$N_l = \frac{1}{k} \sum_{i=1}^k \sqrt{1 - \sigma_i^2}. \quad (7)$$

This is zero for identical subspaces and increases as the leading subspaces rotate apart.

3.5 Diagnostic Analyses

To further characterize residual-stream geometry, targeted diagnostics are performed on selected layers. Activations are captured at multiple stages within a transformer block. Figure 2 illustrates these capture points, separating the stored raw residual state from the normalized states consumed by sublayers and from the outputs written back into the residual pathway.

The same spectral metrics used in the main analysis are computed at each stage. These diagnostics identify where geometric concentration emerges and how it evolves throughout block computation. Table 4 summarizes the diagnostic analyses and their measurements.

Table 4: Summary of diagnostic analyses.

Diagnostic	Purpose	Measurements
Block-stage activations	Localize geometric changes	Spectral metrics at all block stages
Normalization ablations	Test robustness to normalization	Raw, L_2 , RMS, standardized, $-\text{PC}_1$, $-\text{PC}_5$
Blockwise top-PC tracking	Track a fixed raw top-PC direction through blocks	Variance and variance-share along top PC
RMSNorm compression	Measure directional attenuation	Top-PC compression vs. random directions
MoE routing	Compare geometry with routing behavior	Load, entropy, normalized entropy, imbalance

3.5.1 Normalization Ablations

For selected activations, spectra are additionally computed under the transformed views listed in Table 5.

Table 5: Normalization-ablation views.

View	Transformation
Raw	Original activations
L2	Token-wise L_2 normalization
RMS	Token-wise RMS normalization
Standardized	Feature-wise standardization
–PC ₁	Remove leading principal component
–PC ₅	Remove leading five principal components

These ablations distinguish concentration arising from global scale or dominant common directions from concentration that persists under normalization.

3.5.2 Blockwise Top-PC Tracking Under RMSNorm

For RMSNorm-based models [14], the top principal direction of the raw block input is computed and its variance contribution is tracked throughout the block. RMSNorm-associated compression along this direction is additionally compared against randomly sampled control directions. The diagnostic follows the same direction across normalization, attention, MLP, and residual-addition stages; it is observational and does not intervene on activations.

3.5.3 Mixture-of-Experts Routing

For Mixtral-8x7B [18], aggregate routing statistics are additionally collected from MoE layers, including expert load, router entropy, normalized entropy, and load imbalance. These statistics are used only as aggregate routing diagnostics and are not interpreted as direct evidence of human-interpretable expert specialization. They allow residual-stream geometry to be compared against expert-routing behavior and assess whether geometric concentration covaries with simple routing imbalance.

4 Results

4.1 Depth Dependence of Residual Effective Dimension

Residual-stream effective dimensionality varies strongly with depth. The central depth-wise pattern is contraction followed by re-expansion, rather than monotonic collapse. The lowest-dimensional regions appear early or in the early-middle portion of the network, not only at the final layer. Table 6 reports the minimum normalized participation-ratio dimension and associated concentration statistics for each model.

Table 6: Minimum normalized participation-ratio dimension across models. $L_{\min}$ denotes the layer with minimum D_{PR}/d . Top-32 is the cumulative variance share explained by the leading 32 principal components.

Model	$L_{\min}$	Depth	Min D_{PR}/d	k_{90}	Top-32
Mixtral-8x7B	2	0.065	0.000244	1	0.999916
Qwen2.5-14B Base	5	0.106	0.000196	1	0.998848
Qwen2.5-14B Inst.	5	0.106	0.000196	1	0.998846
Qwen2.5-7B Base	5	0.185	0.000280	1	0.998037
Qwen2.5-7B Inst.	5	0.185	0.000280	1	0.998267
GPT-2 Small	3	0.273	0.001721	4	0.936675
GPT-2 Medium	4	0.174	0.001493	5	0.927342
Pythia-1.4B	5	0.217	0.000894	353	0.805600

Effective dimension does not decrease monotonically across depth. For Qwen2.5-14B Base, the final normalized effective dimension is $44.75\times$ larger than its minimum; for Qwen2.5-14B Instruct, the ratio is $37.78\times$; and for Mixtral-8x7B, it is $37.12\times$. Thus, the lowest-dimensional layer is not representative of the final representation state. This matters because a small minimum alone is not enough to identify a near-rank-1 covariance regime: Pythia has a low-dimensional hotspot, but its $k_{90} = 353$ indicates a broader subspace rather than a single top-PC-dominated direction. Figure 3 summarizes the full depth profiles, near-rank-1 intervals, and model-level statistics across the suite:

Figure 3a shows the layerwise contraction and re-expansion, while Figure 3b compares model-level averages and subspace novelty.

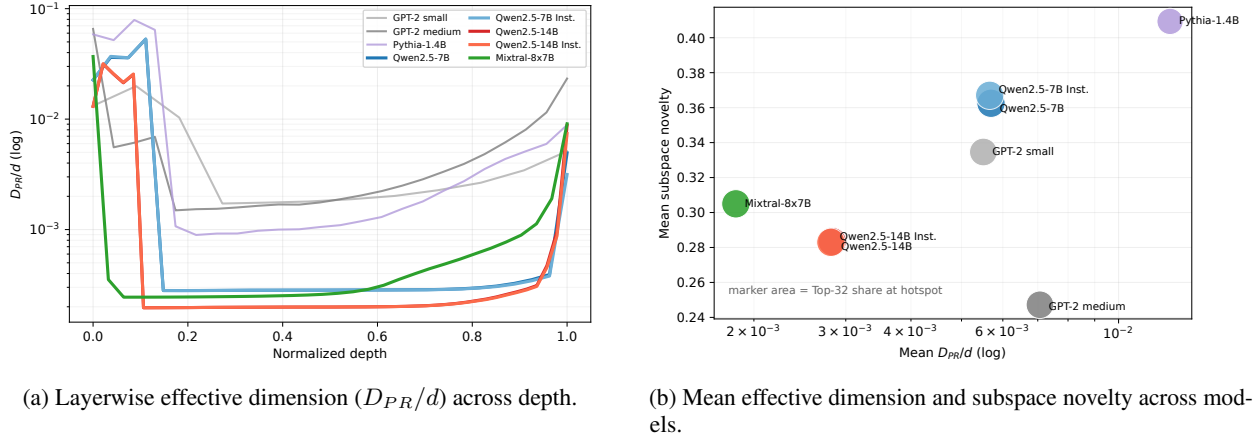


Figure 3: Residual-stream geometry across the evaluated model suite.

4.2 Sustained Low-Dimensional Regimes in Qwen and Mixtral

Under this protocol, Qwen2.5 and Mixtral exhibit the strongest low-dimensional raw residual covariance regimes. Using the operational near-rank-1 covariance criterion from Section 3.4.1, Qwen2.5 and Mixtral contain consecutive near-rank-1 plateaus, whereas GPT-2 and Pythia do not. This criterion describes covariance concentration, not a claim that the model’s computation is rank-1. Table 7 lists the layers satisfying this criterion and the representative cross-layer alignment summaries.

Table 7: Near-rank-1 intervals and sampled raw top-PC alignment. Cosines are absolute values over sampled plateau-layer pairs.

Model	Near-rank-1 layers	Count	Plateau sampled layers	Min	Mean	Max
Qwen2.5-14B Base	5–25	21	L05, L10, L20	0.999708	0.999832	0.999925
Qwen2.5-14B Inst.	5–25	21	L05, L10, L20	0.999691	0.999826	0.999927
Qwen2.5-7B Inst.	4–18	15	not sampled	–	–	–
Qwen2.5-7B Base	4–17	14	L04, L05, L10, L16	0.996757	0.998550	0.999807
Mixtral-8x7B	2–10	9	L02, L03, L04, L05, L10	0.999980	0.999992	1.000000
GPT-2 Small	none	0	–	–	–	–
GPT-2 Medium	none	0	–	–	–	–
Pythia-1.4B	none	0	–	–	–	–

The Qwen2.5-14B variants satisfy the near-rank-1 covariance condition across 21 consecutive layers. The sustained near-rank-1 plateaus appear in the measured RMSNorm-based Qwen2.5 and Mixtral models, but not in the evaluated GPT-2 or Pythia baselines. In contrast, Pythia’s minimum has $k_{90} = 353$, indicating a broader low-dimensional subspace rather than the same top-PC-dominated regime.

4.3 Cross-Layer Alignment of Dominant Principal Components

Cross-layer alignment analysis tests whether near-rank-1 covariance layers share a common leading direction. In Qwen2.5-14B Base and Instruct, sampled plateau-layer top-PC cosine means are 0.999832 and 0.999826, respectively. Qwen2.5-7B Base has mean sampled plateau alignment 0.998550, and Mixtral has mean sampled plateau alignment 0.999992. The alignment columns in Table 7 give the corresponding minimum, mean, and maximum cosines.

The observed cross-layer top-PC alignment suggests a persistent common direction in the raw residual stream. This alignment indicates persistence of the dominant covariance direction, but does not by itself identify the semantic or causal role of that direction.

4.4 Comparison of Raw and Normalized Residual Geometry

Raw near-rank-1 covariance does not necessarily imply rank-1 computation. The next test is therefore whether the low-dimensional raw residual state remains low-dimensional under the normalized views that sublayers actually consume. When token-wise L_2 normalization, token-wise RMS normalization, or RMSNorm-normalized sublayer inputs are examined, the effective dimension increases substantially. The PC-removal views test whether the apparent low dimensionality is dominated by one or a few common directions. Figure 4 summarizes the raw, normalized, and PC-removal comparisons across models.

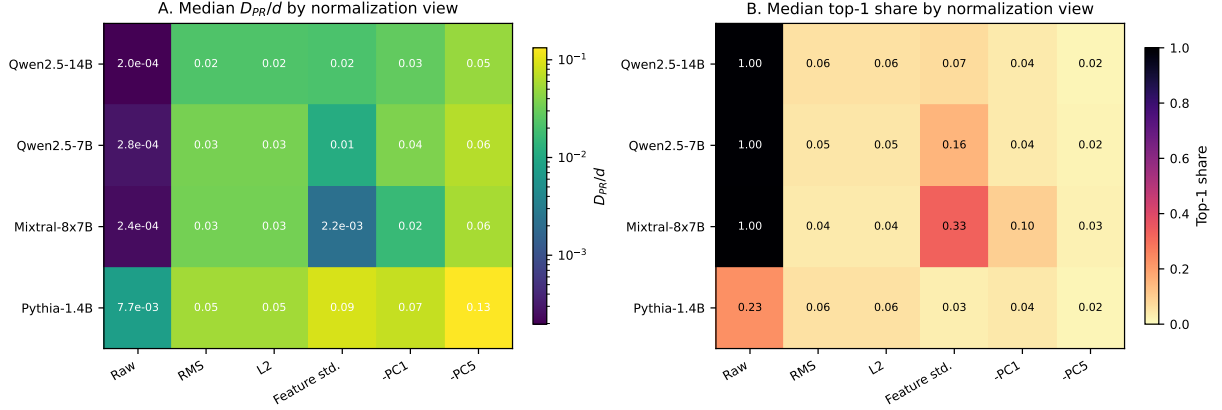


Figure 4: Normalization-ablation summary across models and variants. Raw Qwen2.5 and Mixtral residual streams have very low effective dimension and near-unit top-1 share, but token-wise normalization and PC removal produce higher-dimensional covariance spectra. Pythia provides a contrast case without the same sustained low-dimensional raw regime.

Across sampled Qwen2.5-14B plateau layers, median D_{PR}/d increases from 1.99×10^{-4} in the raw view to 2.48×10^{-2} after RMS normalization, while median top-1 share decreases from 0.9983 to 0.0661. Mixtral shows the same qualitative shift: median raw D_{PR}/d is 2.44×10^{-4} with median top-1 share 0.9994, whereas RMS normalization gives median $D_{PR}/d = 3.40 \times 10^{-2}$ and median top-1 share 0.0418.

RMSNorm directional compression measures variance reduction along the leading raw component. Values below one indicate that RMSNorm reduces variance along the tested direction. For a raw activation h and its raw top principal component v , the diagnostic computes

$$\rho_v = \frac{\text{Var}(\text{RMSNorm}(h)v)}{\text{Var}(hv)} \quad (8)$$

and compare it against compression along random directions. Random directions control for the overall scale change induced by RMSNorm. Table 8 reports the measured top-PC and random-direction compression ratios.

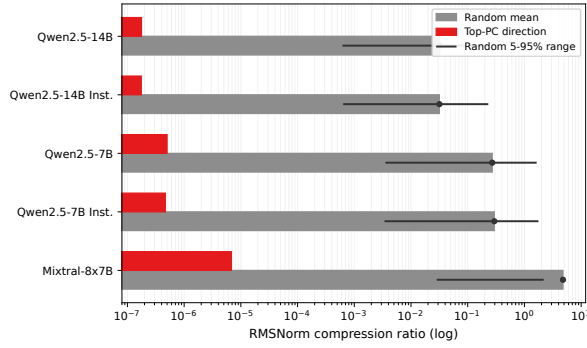
Table 8: RMSNorm compression of the raw top-PC direction compared with random directions.

Model	Layer	Top-PC compression	Random mean compression	Specificity
Qwen2.5-14B Base	5	1.77×10^{-7}	3.18×10^{-2}	5.55×10^{-6}
Qwen2.5-14B Inst.	5	1.76×10^{-7}	3.17×10^{-2}	5.54×10^{-6}
Qwen2.5-7B Base	4	5.00×10^{-7}	2.71×10^{-1}	1.85×10^{-6}
Qwen2.5-7B Inst.	4	4.66×10^{-7}	2.94×10^{-1}	1.58×10^{-6}
Mixtral-8x7B	2	6.82×10^{-6}	4.78	1.43×10^{-6}

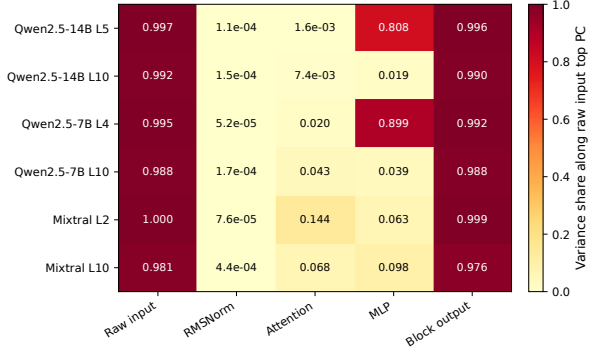
The top-PC direction is attenuated by orders of magnitude more than typical random directions. Specificity is the top-PC compression ratio divided by the random-direction mean; smaller specificity values indicate more selective attenuation of the top-PC direction. In Mixtral, the random-direction mean is greater than one, showing that RMSNorm can increase variance along some directions while still sharply decreasing variance along the raw top-PC direction.

4.5 Block-Stage Evolution Under RMSNorm

Stage-wise measurements reveal a consistent pattern across block stages. The dominant direction is large in the raw residual stream, nearly absent as a variance component after RMSNorm, and large again after residual-writing stages. Figure 5 summarizes this diagnostic: Figure 5a compares top-PC compression against random-direction controls, while Figure 5b tracks the same raw-input direction across block stages. Table 9 reports the corresponding variance shares for sampled layers.



(a) RMSNorm compression rate for the raw top-PC direction.



(b) Variance share along the raw-input top PC across block stages.

Figure 5: RMSNorm diagnostic summary. The top-PC direction has a much smaller compression ratio than random directions, and stage-wise measurements show lower variance share along the same direction after RMSNorm with high variance share again at the block output.

Table 9: Variance share along the raw-input top principal component across block stages.

Model/layer	Raw input	RMSNorm input	Attention output	MLP output	Block output
Qwen2.5-14B L5	0.9967	0.000114	0.001614	0.8083	0.9960
Qwen2.5-14B L10	0.9917	0.000147	0.007436	0.0187	0.9897
Qwen2.5-7B L4	0.9947	0.000052	0.02004	0.8987	0.9923
Qwen2.5-7B L10	0.9884	0.000167	0.04322	0.0393	0.9876
Mixtral-8x7B L2	0.9997	0.000076	0.1440	0.0632	0.9993
Mixtral-8x7B L10	0.9813	0.000438	0.0684	0.0975	0.9755

In these sampled layers, the raw-input top-PC direction explains most raw input variance. After RMSNorm, the same direction explains little of the normalized variance. By the block output, its variance share is again high. These measurements do not identify a complete causal mechanism; they localize where the variance share is low or high under this observational diagnostic.

4.6 Block-Stage Localization of Low-Dimensional Onset

Block-localization diagnostics around the onset of low effective dimension show that attention outputs and RMSNorm-normalized inputs remain comparatively high-dimensional, while MLP outputs and block outputs satisfy the near-rank-1 criterion. Table 10 gives the onset measurements for Qwen2.5 Base models.

These measurements localize the emergence of the leading component to MLP and block outputs, while attention outputs and normalized inputs remain comparatively high-dimensional at the onset layers.

4.7 Controls, Contrast Cases, and Metric Agreement

The main spectral metrics agree directionally. Across layers and models, D_{PR}/d correlates with entropy effective rank at 0.967, with top-1 variance share at -0.745 , and with top-32 variance share at -0.764 . This agreement reduces the likelihood that the observations depend on a single dimensionality measure.

Under this protocol, instruction-tuned and base Qwen2.5 pairs have nearly identical measured geometry: the mean absolute D_{PR}/d difference is 9.92×10^{-5} for the 7B pair and 4.65×10^{-5} for the 14B pair, with the same layer-5

Table 10: Block-localization diagnostics around onset.

Capture	D_{PR}/d	Top-1	k_{90}
14B L04 attention	0.02554	0.0434	499
14B L04 input norm	0.01511	0.0604	412
14B L04 MLP output	0.000195	0.9997	1
14B L04 block output	0.000196	0.9983	1
14B L05 input	0.000196	0.9983	1
7B L03 attention	0.03806	0.0386	439
7B L03 input norm	0.02811	0.0533	426
7B L03 MLP output	0.000280	0.9991	1
7B L03 block output	0.000281	0.9971	1
7B L04 input	0.000281	0.9971	1

hotspot in both pairs. This is consistent with the measured middle-layer raw residual covariance being shared between each base model and its instruction-tuned counterpart.

Mixtral provides a sparse MoE control. It exhibits a strong low-dimensional raw residual regime, but its aggregate router statistics do not show a simple expert-load concentration: mean normalized router entropy is 0.995, minimum normalized router entropy is 0.975, mean load imbalance is 1.191, and maximum load imbalance is 1.418. Correlations between residual geometry and router statistics also do not by themselves account for the phenomenon: D_{PR}/d correlates with normalized router entropy at -0.261 and with load imbalance at 0.580 , while top-32 variance share correlates with normalized router entropy at -0.037 .

The phenomenon is not observed uniformly across the evaluated models. GPT-2 Small and GPT-2 Medium have low-dimensional hotspots but no sustained near-rank-1 plateau. Pythia-1.4B has the highest mean normalized effective dimension among the evaluated models (0.012551) and the highest mean feature-subspace novelty (0.409332), and its hotspot remains qualitatively distinct from Qwen2.5 and Mixtral. The sustained near-rank-1 plateaus are specific to the measured RMSNorm-based architectures, especially Qwen2.5 and Mixtral; GPT-2 and Pythia serve as contrast cases.

5 Discussion

The central finding is a separation between raw residual storage geometry and normalized computation-facing geometry. In Qwen2.5 and Mixtral, raw residual covariance can be dominated by a single highly aligned direction across many middle layers, yet the normalized inputs consumed by attention and MLP sublayers remain substantially higher-dimensional. The same model state can therefore look low-dimensional as a stored residual vector while presenting richer geometry to the next computation step.

Covariance rank and computational rank describe different aspects of representation structure. Covariance rank summarizes how variance is distributed across raw residual vectors in a token sample. Computational rank would require a statement about the effective degrees of freedom used by downstream sublayers or by the model’s input-output behavior. A model can therefore carry a large common-mode direction in the raw residual stream while the state consumed by attention and MLP sublayers remains substantially higher-dimensional.

For mechanistic interpretability, spectral outliers should be treated as hypotheses rather than mechanisms. A leading principal component may reflect a storage convention, a mostly inert common-mode direction, or a functional direction used by later computation; the present measurements do not distinguish those cases by themselves. Reporting both raw and normalized spectra helps separate storage geometry from computation-facing geometry and reduces the risk of describing a model as representationally collapsed when the relevant sublayer inputs remain high-dimensional.

RMSNorm complicates residual-spectrum interpretation because pre-normalized transformer blocks use the raw residual stream both as a storage pathway and as the source from which normalized sublayer inputs are computed. In the measured RMSNorm layers, the leading raw component has much lower variance share after normalization and can later have high variance share again at MLP or block outputs. This pattern is consistent with selective attenuation of the top-PC direction under this diagnostic, but it does not by itself show why later block stages re-express high variance along that direction.

Several observations help explain why Qwen2.5 can show an especially strong near-rank-1 raw covariance plateau. First, residual addition allows a common-mode direction to accumulate across blocks even if sublayers read a normalized version of the state. Second, RMSNorm can sharply reduce variance along that same raw direction before attention and MLP computation, making the direction large in storage but small in the immediate computation-facing input. Third, the

onset diagnostics localize the first strong spike to MLP and block-output stages in the sampled Qwen layers, suggesting that residual-writing stages can reintroduce the dominant raw direction after normalization. Fourth, the high cross-layer cosine similarity indicates that later plateau layers are not producing unrelated spikes; they largely preserve or rewrite a shared leading direction. These explanations remain partial. The current data support a storage-normalization cycle involving residual accumulation, normalization interaction, and block-output rewriting, but they do not prove that MLP expansion, common-mode bookkeeping, or architectural depth is individually sufficient to cause the plateau.

Weight-space diagnostics involving the MLP down-projection (W_{down}) are mixed and therefore do not provide a single explanation for the plateau. Qwen2.5-7B shows high alignment after the onset of the low-dimensional regime, Qwen2.5-14B is weaker at onset and only moderate later, and Pythia can show high alignment without a Qwen-like plateau. These measurements are best interpreted as suggestive architecture-level correlates, not as evidence that the down-projection alone causes the observed residual geometry.

Relative to prior anisotropy work, the results are consistent with the broader finding that dominant common components can appear in neural language representations [2, 3]. The new point is narrower and architecture-specific: in the evaluated pre-normalized RMSNorm models, the raw residual common direction can be nearly rank-1 over a sustained depth interval while the normalized sublayer input remains much higher-dimensional. Compared with effective-rank diagnostics that summarize a single representation view [10, 11], the block-stage measurements show that the conclusion depends strongly on which internal view is measured. Compared with mechanistic-interpretability studies of residual features and circuits [4, 5, 6, 7], this paper does not identify a semantic circuit; it supplies a covariance-level warning that raw residual spectra and computation-facing spectra can tell different stories.

The unresolved question is whether the leading raw component is functionally load-bearing or mostly inert common-mode bookkeeping. The most direct next experiment is causal: project away the top principal direction at selected layers and measure next-token loss, logit KL divergence, and top-token agreement against random-direction controls.

6 Limitations

The study uses WikiText-103 validation text, a fixed frozen-evaluation protocol, and a limited model suite. Models are evaluated with their native tokenizers and unequal token budgets, so the results should be interpreted as protocol-level measurements rather than universal statements about all prompts, datasets, tokenizations, or scales. GPT-2, Pythia, Qwen2.5, and Mixtral provide useful contrasts, but they do not exhaust the design space of transformer architectures.

The analysis is descriptive rather than causal. Covariance spectra, compression ratios, block-stage measurements, and routing summaries identify where a variance-dominant direction appears or is attenuated, but they do not establish that the direction is necessary for prediction. Activation patching, direction removal during inference, loss measurements, and logit-divergence tests are not performed in this work.

Principal components are also not semantic objects by default. A top-PC direction may combine many features, reflect a common-mode offset, or encode information in a form that is difficult to interpret from covariance alone. The block-localization measurements associate the onset with MLP and block outputs, but they do not explain why those stages produce high variance along the leading direction. Weight-level analysis, causal interventions, and training-dynamics studies are needed to determine whether the observed geometry is a mechanism, a byproduct, or a mixture of both.

7 Conclusion

This study analyzes residual-stream geometry in pretrained language models using frozen activations, covariance spectra, and block-stage diagnostics. The main empirical pattern is that Qwen2.5 and Mixtral exhibit sustained low-dimensional raw residual covariance regimes with cross-layer-aligned leading principal directions, while GPT-2 and Pythia provide contrast cases without the same sustained plateau. However, the low-dimensional raw spectrum is not the whole computational story. RMSNorm-normalized sublayer inputs remain much higher-dimensional, and the top raw direction contributes little variance immediately after normalization even when it dominates the stored residual stream again at later block stages.

These results support a narrower interpretation of residual-stream anisotropy. A dominant raw covariance direction is an important diagnostic signal, but it does not by itself imply that the model’s computation has collapsed to one dimension. The next step is causal: remove or perturb the leading direction and measure effects on loss, logits, and token predictions. Residual-stream spectra are therefore valuable diagnostics, but they should be interpreted alongside normalization-facing states before drawing conclusions about computational dimensionality.

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